

Experiment 7.2

Fourier Series of Half-Wave Rectifier Output

Program:

% Fourier Series of Half-Wave Rectifier Output

clc;

clear;

close all;

% Time axis

t = 0:0.001:2*pi;

% Half-wave rectified signal

x = sin(t);

x(x < 0) = 0;

% Fourier Series approximation

N = 10; % Number of harmonics

a0 = (1/pi)*trapz(t,x);

fs_signal = a0/2;

for n = 1:N

an = (1/pi)*trapz(t,x.*cos(n*t));

bn = (1/pi)*trapz(t,x.*sin(n*t));

fs_signal = fs_signal + an*cos(n*t) + bn*sin(n*t);

end

% Plotting

figure;

subplot(2,1,1);

plot(t,x,'LineWidth',2);

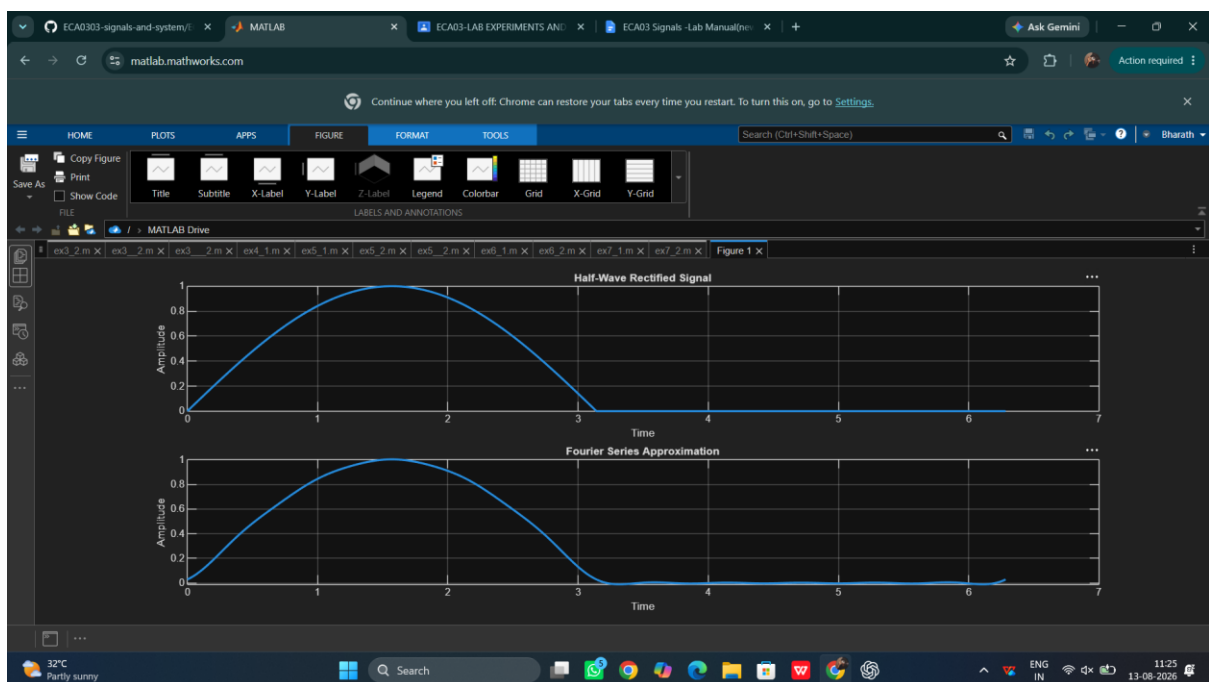
grid on;

```

title('Half-Wave Rectified Signal');
xlabel('Time');
ylabel('Amplitude');
subplot(2,1,2);
plot(t,fs_signal,'LineWidth',2);
grid on;
title('Fourier Series Approximation');
xlabel('Time');
ylabel('Amplitude');

```

Output Waveform:



Result:

The Fourier Series of the half-wave rectified signal was obtained using MATLAB. The reconstructed waveform using Fourier coefficients closely matched the original half-wave rectified signal, demonstrating the Fourier Series representation of periodic signals.